PROBABILITY DICE & MODULAR SPINNER PACK

Random Variables, Sample Spaces & Statistical Distributions

SERIES 1 • CARD 07 / 12 • DOMAIN: SAMPLE SPACES & PROBABILITY DISTRIBUTIONS

CARD 07 / 12 • PROB-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

SAMPLE SPACES, 2d6 SUMS & NORMAL CURVES

Randomness follows strict combinatorial laws.

Rolling two fair 6-sided dice gives 36 outcomes:

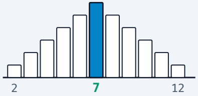
Sample Space = 6 × 6 = 36 Pair Outcomes

2d6 sum forms a triangle distribution peak at 7!

CALIBRATED 16.00 MM FAIR DICE DATUM

- Fair d6: 16.00 mm precision cube, balanced pips
- Polyhedral d20: 20-sided icosahedron (1/20 odds)
- Low-Friction Spinner: Precision brass bushing pivot

2d6 SUM PROBABILITY: TRIANGULAR PEAK AT 7 (6/



PEAK AT 7 (16.7%)

6 / 36 Combinations

Symmetric Triangle

Sample Space: 6×6=3

Birth of the Bell Curve

2. KIT PARTS INVENTORY (14 PIECES)

COMPONENT	QTY	SPEC	COLOR
Fair d6 Dice	4x	16.0 mm Fair	High-Contrast
Polyhedral d20	1x	20-Sided d20	Emerald Green
Precision Hub	1x	Brass Hub	High-Spin Brass
Spinner Dials	4x	3 Dials Set	Polymer Plates
Sample Grid Lid	1x	6×6 Matrix	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal

TOTAL: 10 Working Parts + 2 Extra Dials + Storage Bin + Lid = 14 Parts

PROBABILITY DISTRIBUTION PALETTE

- 4 Color-Separated d6 Dice: Enables independent event tracking.
- Emerald d20 Die: High-visibility prime distribution testing.
- High-Spin Brass Bushing: Ultra-low friction for random spins.
- Clear Calibration: High-contrast legible numerals.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Seat the 4 d6 dice and 1 d20 into bottom bays. Stack spinner dials flat and click brass hub onto post. Empty bays show missing pieces instantly. Snap lid shut.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

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Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

The sum of independent random variables tends toward a normal distribution as sample size grows.

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LESSON GUIDE: 2d6 SAMPLE SPACES & INDEPENDENT EVENTS

Step-by-Step Hands-On Tutorial: Why 7 is the Most Likely Roll (P = 6/36)

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LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/dice-and-spinner/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 7.SP.C.8 (Compound Events)
- CCSS HSS.CP.A.2 (Independence)
- TEKS 111.27.b.6.B (Representati...)

Grades 5–10 Probability & Statistics

💡 GUIDING INQUIRY

Why are there six different combinations to roll a 7, but only one combination to roll a 2?

① STEP 1: INSPECT THE 6 × 6 SAMPLE SPACE

Examine the 6 × 6 grid on the lid stage.
Die 1 rows (1-6) × Die 2 columns (1-6) =
36 total outcomes.
Every individual pair (like [3,4] or [6,1]) has probability 1/36.

② STEP 2: COUNT PATHS TO EACH SUM

Trace diagonals of constant sum across the grid:

- Sum 2: Only 1 way [1,1] → P = 1/36 (2.8%)
- Sum 7: 6 ways [1,6],[2,5],[3,4],[4,3],[5,2],[6,1] → P =

3–4 STEPS 3 & 4: SAMPLE TRIALS & CALCULATE

3. SAMPLE EXPERIMENTAL RUNS: Roll 2d6 three times: 7, 4, 10.

4. COMPUTE SAMPLE MEAN: Calculate mean on calculator:

Press Dice & Statistics Icon ([🎲], [Σ+] & [g][x̄]) and run [g] [CLΣ] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g] [x̄] → 7.

Independent Event: d6=6 and Spinner=10 → 1 [ENTER] 6 [÷] 1 [ENTER] 10 [÷] [×] [FDISP] → 1/60

⑤ CORE PRINCIPLE: THE BIRTH OF THE BELL CURVE

Single dice have flat uniform distributions (each face is 1/6). But when two dice add together,
extreme sums require extreme coordination, while central sums have many different additive combinations!
This combinatorial clustering is the foundation of the normal distribution Bell Curve!

★ EXTENSION INVESTIGATION: EVEN SUM PROBABILITY

Count all 2d6 combinations that give an EVEN sum (2, 4, 6, 8, 10, 12). Exactly 18 out of 36 outcomes! P = 50.0%! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

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